

CP020003

ARTIFICIAL INTELLIGENCE



Khon Kaen University · 2026

Week 12

# Generative AI & the modern AI

From describing the world to inventing it



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[https://github.com/kaopanboonyuen/CP020003\\_ArtificialIntelligence\\_2026s1](https://github.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1)



# Today's map

Seven ideas, one thread: making instead of describing.

1

## Two minds, two ways

LLM vs. diffusion

2

## Text → image

Stable Diffusion v1.5

3

## Image → text

Captioning with BLIP

4

## The round trip

How much meaning survives

5

## Inpainting

Editing only part of an image

6

## One-step “turbo” generation

Distillation makes it instant

7

## The 2026 frontier

What's newer than diffusion

## 1

# Two minds, two ways to generate

Same goal — make something new — completely different process.

## Large language model



committed → committed → committed → writing now ...

*One token at a time, left to right, never undone.*

**A large language model is autoregressive.** It picks one token, appends it, and conditions the next pick on everything before it — it can never reach back and change a word it already committed to.

## Diffusion model



*The whole canvas, revised together, many passes.*

**A diffusion model works the opposite way.** It starts from pure noise and, at every step, revises the entire image at once — nothing is final until the last pass.

### From the notebook — Section 1

We load Qwen2.5-0.5B-Instruct and print each token the instant it's generated — students watch the commitment happen live.

# LLMs vs. diffusion, side by side

|                  | Large language models                                 | Diffusion models                                     |
|------------------|-------------------------------------------------------|------------------------------------------------------|
| Data type        | Text — discrete tokens                                | Images / audio — continuous signal                   |
| How it generates | One token at a time, left to right                    | Whole canvas, refined over many passes               |
| Training signal  | Predict the next token                                | Predict the noise that was added                     |
| Analogy          | Writing a sentence word by word, committing as you go | Sculpting a statue from noise, pass after pass       |
| 2026 frontier    | Longer context, tool use, reasoning chains            | Fewer steps (distillation), new math (flow matching) |

## 2

## Text → image: turning words into pixels

The flagship demo of generative AI — the one every student remembers.

Given a text prompt, generate a brand-new image that matches it. We run this with Stable Diffusion v1.5, guided by a text encoder that turns your words into the signal the denoiser follows at every step.

### Two knobs control almost every generation

S

#### **num\_inference\_steps**

How many denoising passes it gets. More steps = more refinement, with diminishing returns and linear extra compute.

G

#### **guidance\_scale**

How strictly it follows your prompt vs. its own sense of a plausible image. Too low ignores you; too high over-sharpens.

#### From the notebook — Section 2.1 & 2.2

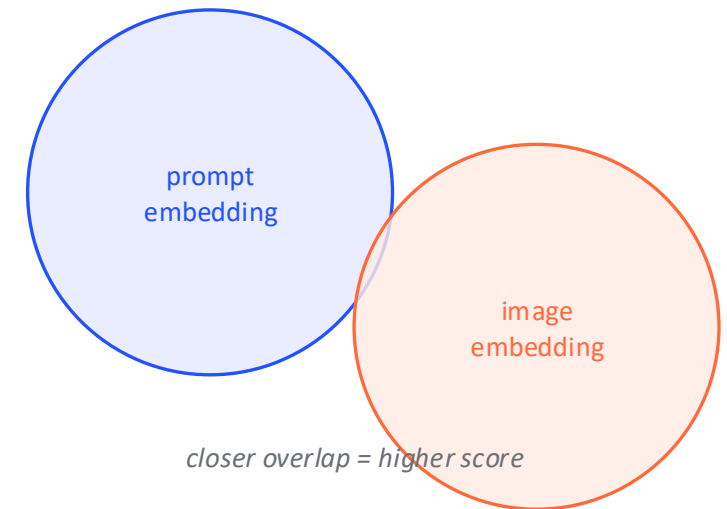
Three prompts, one seed — “a majestic red panda astronaut floating in space, digital art” and two more. Then a full sweep: steps [5, 15, 30] × guidance [1.5, 7.5, 15.0], nine images, one grid.

# Scoring a “vibe” with an actual number

Looking at a grid of images and saying “that one looks better” isn't a metric — it's a feeling. CLIP Score encodes the prompt and the image into the same embedding space and measures how close they land: a real, comparable number.

**openai/clip-vit-base-patch32**

The exact checkpoint we use to score every image generated this week.



## 3 Image → text: describing what we made

The reverse direction — and the backbone of alt-text, search, and accessibility tools.

Given an image, generate a caption in natural language. We use BLIP, a small, fast captioning model, and point it straight at the art the class just generated — closing the loop between what diffusion made and how a language model reads it back.

### From the notebook — Section 3

Salesforce/blip-image-captioning-base runs on every image from Section 2, turning pixels back into a sentence a language model can use.

1

### Accessibility

Alt text for screen readers, generated automatically.

2

### Search

Captions become searchable metadata for image libraries.

3

### Moderation

A text summary a downstream system can reason about.

## 4

# The round trip: image → text → image

A game that quietly teaches a deep lesson about meaning and compression.



## Then we compare

Start with a prompt → Image A. Caption it → Prompt B. Regenerate from Prompt B → Image C. Compare A and C.

## What usually survives

The main subject, the setting, and the broad style — a red panda astronaut stays a red panda astronaut.

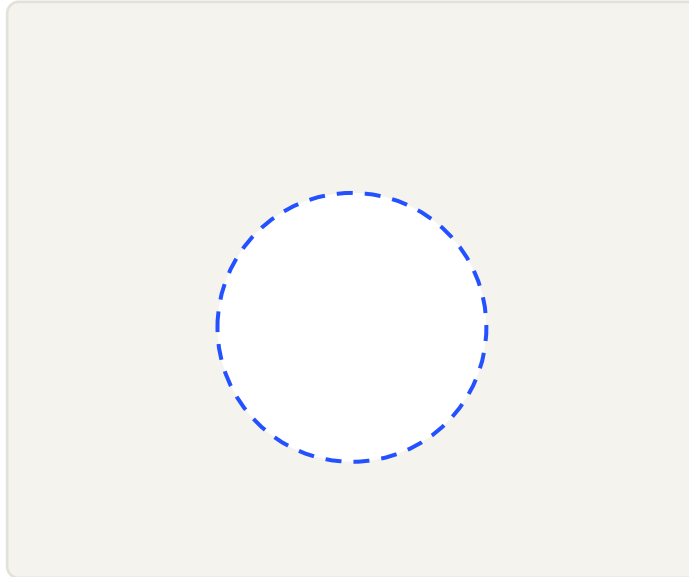
## What usually drifts

Fine detail, exact composition, and mood — every translation between modalities is lossy, and here you can see it happen.

## 5

# Inpainting: editing only part of an image

The technology behind “Magic Eraser” and “Generative Fill.”



*white = edit this*  
*black = leave alone*

Given an image, a mask, and a new prompt, the model regenerates only the masked region while leaving everything outside it untouched — via `StableDiffusionInpaintPipeline`.

**A good inpaint has to pass two checks at once:**



## CLIP score inside the mask

High similarity to the new prompt, measured only on the cropped region.



## MSE outside the mask

Near zero difference from the original — stay quiet everywhere else.

**From the notebook — Section 5.1**

Match inside the circle, stay silent outside it — that's the whole job of an inpainting model.

## 6 Beyond classic diffusion: one-step “turbo” generation

The single biggest practical trend in generative AI since 2024.

Section 2 needed 25–30 denoising passes — the whole reason diffusion feels slow next to a single forward pass through a classifier. Distillation trains a fast student to jump straight to what a slow teacher reaches after dozens of steps.

A

### Adversarial Diffusion Distillation

Powers stabilityai/sd-turbo. A discriminator pushes the few-step student toward the many-step teacher’s output.

L

### Latent Consistency Models

Trained so any point on the noise-to-image path maps to the same final image — letting you leap most of the way there.

20–50 steps



1–4 steps

*the difference between “wait ten seconds” and “instant.”*

7

# The 2026 frontier: what's newer than diffusion

This is where the LLM column and the diffusion column from Slide 4 start to merge.

1

## Flow matching straightens the path

Stable Diffusion 3 and Flux train the model to follow a straight line from noise to image — fewer steps needed, baked in from training instead of bolted on after.

2

## The U-Net retires, the transformer takes over

Diffusion Transformers (DiT) — the backbone behind Sora-style video models — replace the old convolutional denoiser with the same block that powers every LLM.

3

## The autoregressive comeback

Some of the newest multimodal systems generate images token by token, left to right, inside the same transformer that writes the reply — no separate denoiser at all.

### The big picture

2020: generative AI meant GANs. 2023: it meant diffusion. 2026: the frontier is unified multimodal transformers, where “LLM” and “image generator” are two settings of the same machine.

# Wrap-up: the map of modern generative AI

Every task we ran this week, in one table.

| Task              | Model                 | Hugging Face ID                                                | Output             | Key metric            |
|-------------------|-----------------------|----------------------------------------------------------------|--------------------|-----------------------|
| Text generation   | Qwen2.5-0.5B-Instruct | <code>Qwen/Qwen2.5-0.5B-Instruct</code>                        | Token sequence     | Human judgment        |
| Text → image      | Stable Diffusion v1.5 | <code>stable-diffusion-v1-5/stable-diffusion-v1-5</code>       | 512×512 image      | CLIP score            |
| Image → text      | BLIP (base)           | <code>Salesforce/blip-image-captioning-base</code>             | Caption            | Human judgment        |
| Masked editing    | SD v1.5 Inpainting    | <code>stable-diffusion-v1-5/stable-diffusion-inpainting</code> | Edited region      | CLIP (in) / MSE (out) |
| Fast text → image | SD-Turbo              | <code>stabilityai/sd-turbo</code>                              | 512×512, 1–4 steps | CLIP vs. seconds      |



# The AI community building the future.

The platform where the machine learning community collaborates on models, datasets, and applications.

[Explore AI Apps](#)

or

[Browse 2M+ models](#)
[Main](#)
[Tasks](#)
[Libraries](#)
[Languages](#)
[Licenses](#)
[Other](#)

Models 2,555,000

Tasks

 Text Generation

 Any-to-Any

 Image-Text-to-Text

 Image-to-Text

 Image-to-Image

 Text-to-Image

 Text-to-Video

 Text-to-Speech

+ 44

Parameters

Libraries

 PyTorch

 TensorFlow

 JAX

 Transformers

 Diffusers

 Safetensors

 ONNX

 GGUF

 Transformers.js

 MLX

Apps

 vLLM

 TGI

 llama.cpp

 MLX LM

 LM Studio

 Ollama

 Jan

+ 7

Inference Providers

 Groq

 Novita

 Cerebras

 SambaNova

 Nscale

 fal

 Hyperbolic

 Together AI

moonshotai/Kimi-K2.5

 Image-Text-to-Text • Updated 2 days ago • 96.2k • 1.42k

openai/gpt-oss-120b

 Text Generation • 120B • Updated Aug 26, 2025 • 2.88M • 1.4k

Lightricks/LTX-2

 Image-to-Video • Updated 14 days ago • 2.72M • 1.41k

microsoft/VoiceVoice-ASR

 Automatic Speech Recognition • 9B • Updated 6 days ago • 175k • 1.5k

nvidia/personalex-7b-v1

 Audio-to-Audio • Updated 4 days ago • 101k • 1.58k

MiniMaxAI/MiniMax-M2.1

 Text Generation • Updated 5 days ago • 84k • 1.24k

Qwen/Qwen3-ASR-0.6B

 Automatic Speech Recognition • 0.9B • Updated 3 days • 11.9k • 128

deepseek-ai/DeepSeek-OCR-2

 Text Generation • 3B • Updated 4 days ago • 18.3k • 621

arcee-ai/Trinity-Large-Preview

 Text Generation • 399B • Updated 5 days ago • 317 • 115

stepfun-ai/Step-3.5-Flash

 Text Generation • 199B • Updated about 1 hour ago • 44 • 114

**Main** Tasks Libraries Languages Licenses Other

#### Tasks

- Text Generation
- Any-to-Any
- Image-Text-to-Text
- Image-to-Text
- Image-to-Image
- Text-to-Image
- Text-to-Video
- Text-to-Speech
- + 44

#### Parameters



#### Libraries

- PyTorch
- TensorFlow
- JAX
- Transformers
- Diffusers
- GGUF
- MLX
- MLX
- Transformers.js
- Safetensors
- + 44

#### Apps

- vLLM
- llama.cpp
- MLX LM
- LM Studio
- Ollama
- Jan
- Draw Things
- + 10

#### Inference Providers

- Groq
- Novita
- Cerebras
- Nscale
- fal
- Together AI
- Fireworks
- Featherless AI
- + 10

**Models** 3,095,695

☐ Base only ☒ Inference Available Sort: Trending

convaiinnovations/laya

Text Classification • 0.4B • Updated 2 days ago • 3.69k

abenzerps/Qwen-Image-2.1-Uncensored-GGUF

Text-to-Image • 7B • Updated 2 days ago • 716k • 1.8k

prism-ml/Ternary-Bonsai-2-27B-gguf

Text Generation • 27B • Updated about 4 hours ago • 3.11M • 2.09k

Altworld/Hemmingway-1

Text Generation • 27B • Updated 3 days ago • 4.98k • 676

XiaomiMiMo/MiMo-V2.6-Pro-RL

Text Generation • 1T • Updated 4 days ago • 42.1k • 499

XiaomiMiMo/MiMo-V2.6-Distill-Qwen-9B

Image-Text-to-Text • 9B • Updated 4 days ago • 6.65k • 477

XiaomiMiMo/MiMo-V2.6-Flash-RL

Text Generation • 311B • Updated 4 days ago • 20.5k • 462

deepseek-ai/DeepSeek-V4.1-Flash

Image-Text-to-Text • 763B • Updated 16 days ago • 621k • 3.75k

nvidia/Nemotron-3-Diarization

Qwen/Qwen-Image-2.1

Text-to-Image • 7B • Updated 5 days ago • 42.5k • 2.32k

XingChen-AGI/Xing4.0-29B-A4B

Text Generation • 31B • Updated 8 days ago • 43k • 1.69k

Comfy-Org/Qwen-Image-2.1

Updated 3 days ago • 3.27M • 736

Edge0/Audio8-ASR-Infinite

Automatic Speech Recognition • 4B • Updated 2 days ago • 2.85k • 580

TaichuAI/ZDTaichu5.0-9B

Image-Text-to-Text • 10B • Updated 6 days ago • 9.5k • 1.05k

Qwen/Qwen3.8-27B

Image-Text-to-Text • 28B • Updated Aug 14 • 6.58M • 16.3k

AlexWortega/openjev

Text Classification • Updated 1 day ago • 578

Lightricks/LTX-2.5

Image-to-Video • Updated 25 days ago • 1.6M • 5.11k

ISTA-DASLab/Qwen3.8-27B-GSQ-RCO-GGUF

https://huggingface.co/convaiinnovations/laya



Main

Tasks

Libraries

Languages

Licenses

Other

## Tasks



Text Generation



Any-to-Any



Image-Text-to-Text



Image-to-Text



Image-to-Image



Text-to-Image



Text-to-Video



Text-to-Speech

+ 44

## Parameters

< 1B

6B

12B

32B

128B

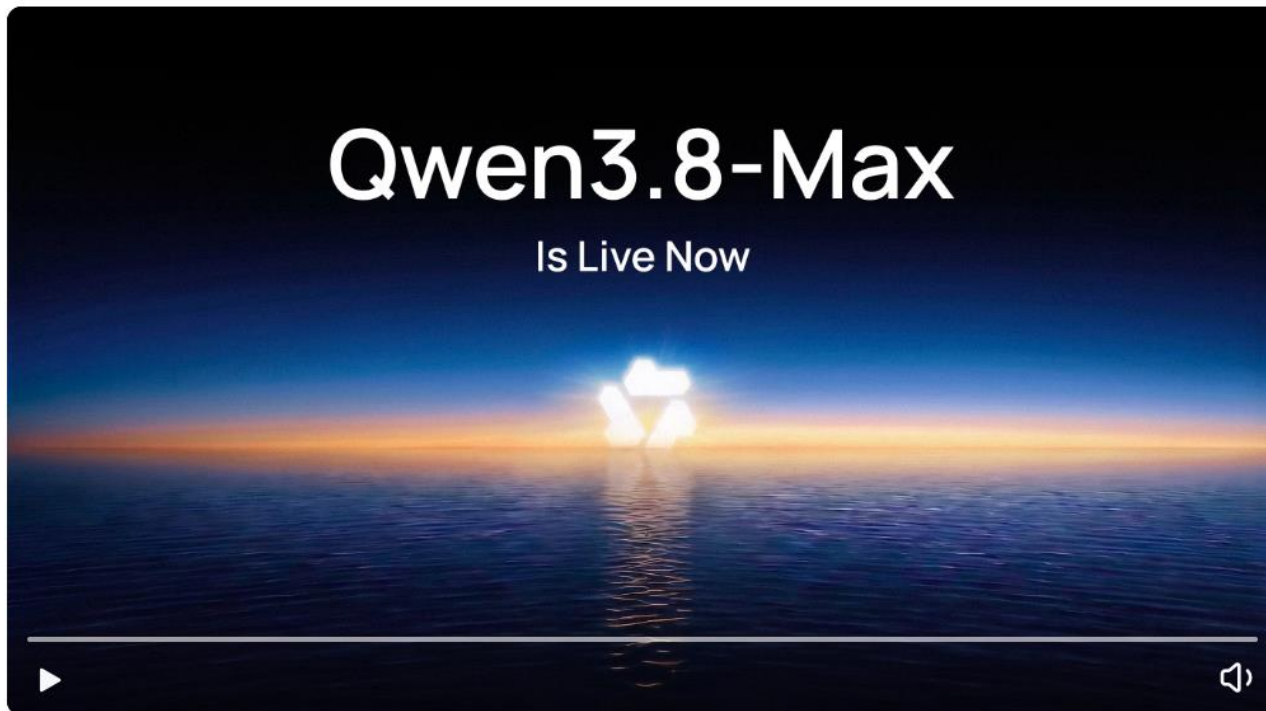
> 500B





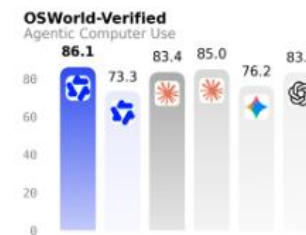
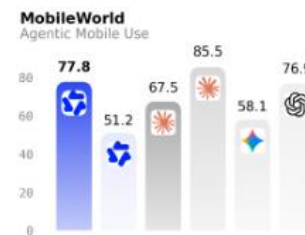
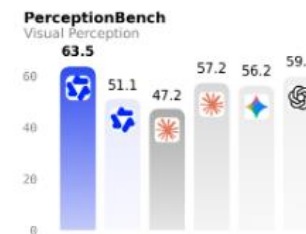
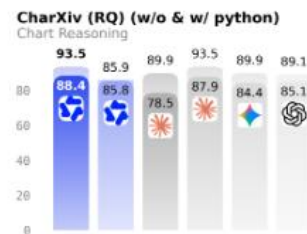
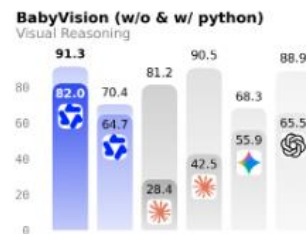
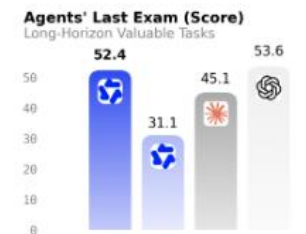
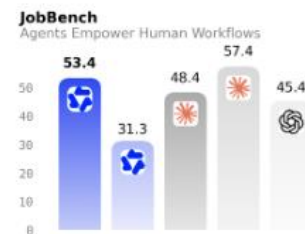
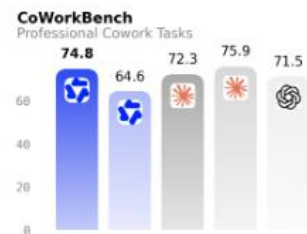
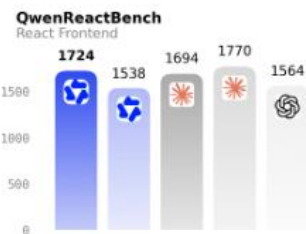
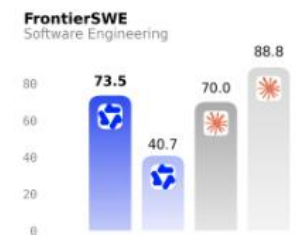
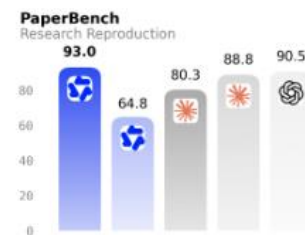
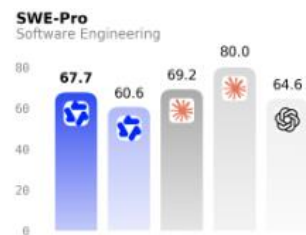
# Qwen3.8-Max

Is Live Now

[QWEN STUDIO ↗](#)[DISCORD ↗](#)

Today, we are officially releasing **Qwen 3.8-Max**, the most capable model in the Qwen family to date. This also marks the first time we will open-source the weights of a Qwen-Max-class model — the open weights will be released next week. Built upon the architectural





■ Qwen 3.8 Max
 ■ Qwen 3.7 Max
 ■ Qwen 3.7 Plus
 ■ Opus4.8
 ■ Fable5
 ■ Gemini3.1-Pro
 ■ GPT5.6 Sol (max)



# Reading AI Model Names

What do "27B", "2.4T", "A95B", "FP8" actually mean?

When people say:

"Qwen3.8 27B"

"Qwen3.8 2.4T-A95B"

they are usually telling you something about the **size and architecture of the model**.

**B = Billion. T = Trillion.**

**B = Billion. T = Trillion.**

**Parameters are the model's learned numbers**

A neural network learns a huge collection of numerical values called **parameters**.

Think of them as:

|  **The model's learned "knobs"**

More parameters generally means **more capacity to represent patterns** — but it does **not** automatically mean a better model.

## The abbreviations

| Symbol | Meaning      |                   |
|--------|--------------|-------------------|
| 1B     | 1 billion    | 1,000,000,000     |
| 27B    | 27 billion   | 27,000,000,000    |
| 100B   | 100 billion  | 100,000,000,000   |
| 1T     | 1 trillion   | 1,000,000,000,000 |
| 2.4T   | 2.4 trillion | 2,400,000,000,000 |

### Easy mental model

B = billion

T = trillion

1T = 1,000B

## So what does “Qwen3.8-27B” mean?

Break the name apart:

### Qwen

The model family / organization

### 3.8

The generation / model version

### 27B

About **27 billion parameters**

So:

Qwen3.8-27B = a Qwen 3.8 model with ~27B parameters

The current Hugging Face listing describes Qwen3.8-27B as an **image-text-to-text** model with about **28B parameters** in its repository metadata.  Hugging Face

# But “bigger = better” is NOT a rule

Imagine two students:

## Student A

🧠 100B parameters

## Student B

🧠 27B parameters

You might expect A to always be better.

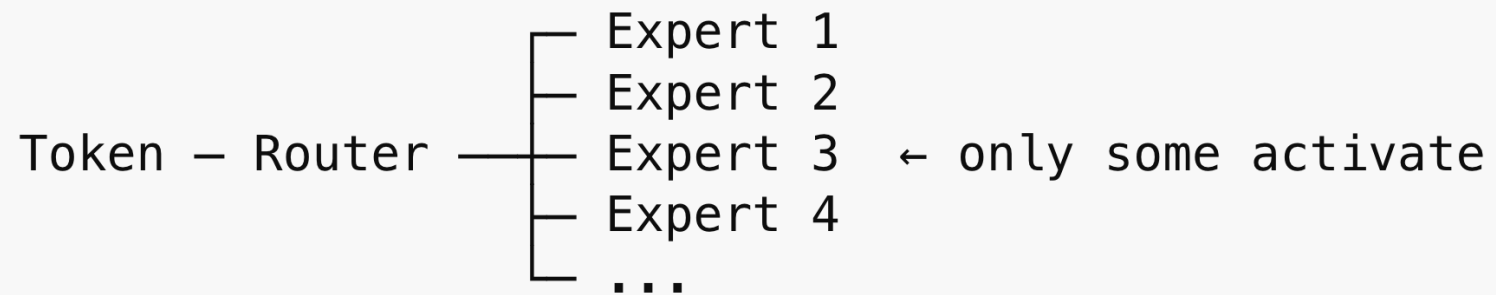
**Not necessarily.**

Model quality also depends on:

- Training data
- Training quality
- Architecture
- Reasoning / post-training
- Context length
- Multimodal capabilities
- Inference strategy
- Quantization
- Tool use
- How the model is evaluated

## Mixture-of-Experts (MoE)

2.4T total parameters



A sparse MoE model can contain a **huge total number of parameters**, while activating only a subset for each token.

## Example: Qwen3.8-2.4T-A95B

Now the name makes much more sense:

### Qwen3.8

Model generation

### 2.4T

2.4 trillion total parameters

### A95B

Approximately **95 billion parameters are activated per token**

So:

**2.4T total ≠ 2.4T active every time**

Qwen's official model card reports **2.4T total parameters and 95B activated parameters**, using a Mixture-of-Experts architecture.  Hugging Face

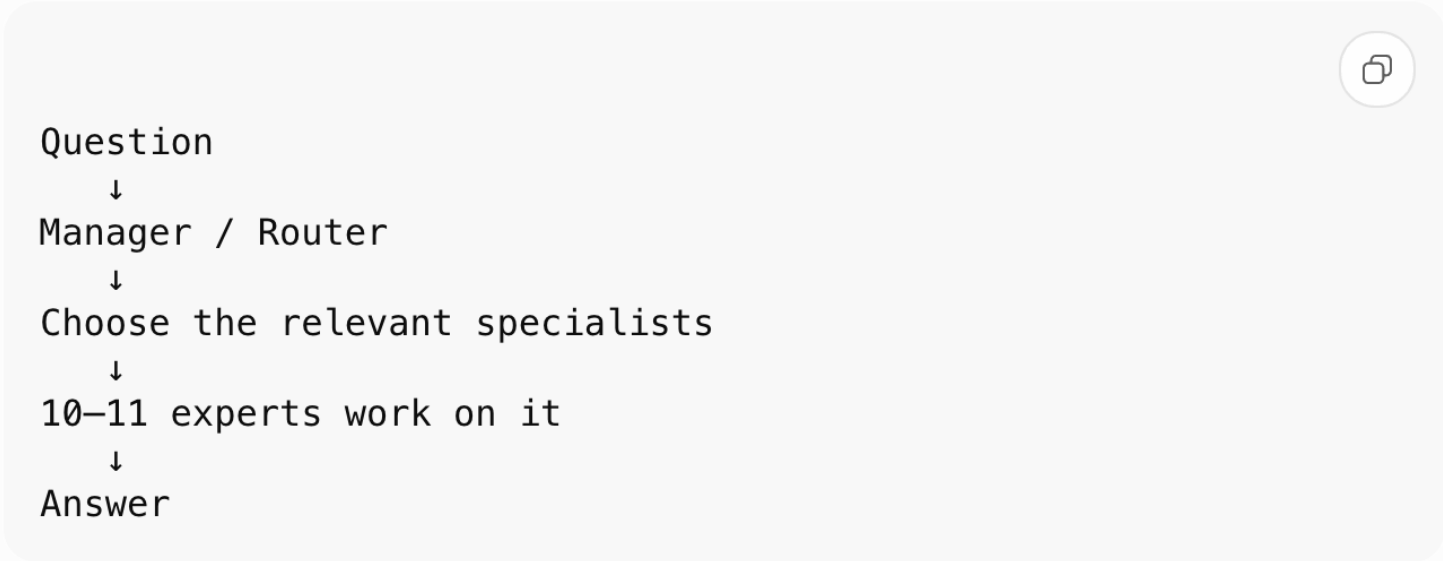
## Think of MoE like a company

Imagine a company has:

👤 **2,400 employees**

But you don't ask **all 2,400 people** to solve every problem.

Instead:



```
graph TD; Question --> Manager[Manager / Router]; Manager --> Specialists[Choose the relevant specialists]; Specialists --> Experts[10-11 experts work on it]; Experts --> Answer
```

Question  
↓  
Manager / Router  
↓  
Choose the relevant specialists  
↓  
10-11 experts work on it  
↓  
Answer

That's roughly the intuition behind **Mixture-of-Experts**.

For Qwen3.8-2.4T-A95B, the model has **512 experts**, with **10 routed experts + 1 shared expert** activated according to the model card. 🤗 Hugging Face

# Model names in 2026: What should you look for?

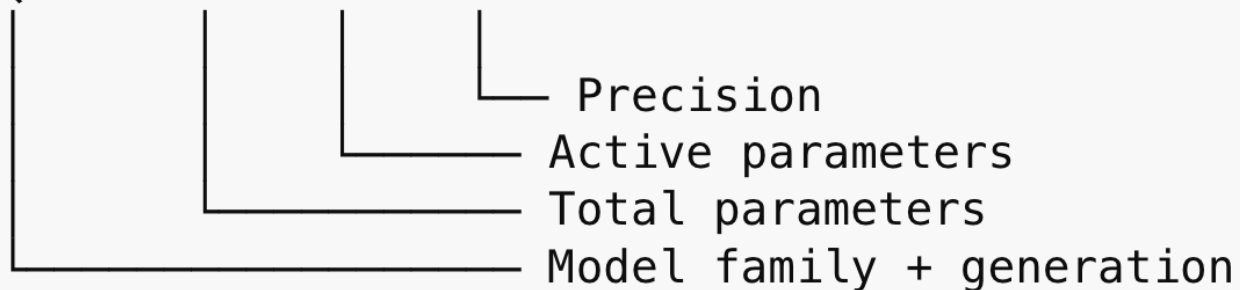
When you see a model name, don't look only at the number.

**Read it as:**

**Family / Version / Size / Architecture / Precision / Purpose**

For example:

Qwen3.8-2.4T-A95B-FP8



# Try it yourself

Four small experiments that turn today's demos into your own results.

1

## Break the guidance scale

Push `guidance_scale` above 20 on your own prompt. Describe the artifacts and connect them back to what the knob does.

2

## Measure your own CLIP score

Write five prompts you think are hard for the model. Generate and score each — which failed lowest, and why?

3

## Run the round trip on your own image

Pick a photo, caption it, regenerate from that caption, and list every detail that changed.

4

## Swap in a newer backbone

Replace SD-Turbo with `sdxl-turbo` or a flow-matching checkpoint and repeat the speed-vs-quality comparison.

# Four ideas to leave the room with

1

## **Discriminative looks; generative makes.**

Everything before this week described the world. Everything this week invents it.

2

## **Autoregressive commits; diffusion revises.**

One never looks back, the other never truly finishes until the last step.

3

## **Speed is a training problem, not just an engineering one.**

Distillation and flow matching teach the model to need fewer steps in the first place.

4

## **The categories are merging.**

The most modern systems don't pick a side of the LLM/diffusion table — they collapse it into one transformer.